# HAMMAD SAJJAD

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## ABOUT ME

Highly motivated and eager to continuously learn and explore new technologies in automation, robotics, and embedded systems. Actively seeking opportunities to enhance technical skills and adapt to emerging engineering tools and industry trends. Committed to applying newly learned knowledge to solve real-world problems and deliver effective solutions.

## SKILLS

**Technical skills**: CAD & CAM |ROS2 (UBUNTU) | MATLAB | Simulink | Automation and Control | MCU Programming (ARM, ATMEL, ESP, PIC)

**Soft Skills**: Team Collaboration, Communication, Project Management, Efficiency, Decision-Making, Adaptability, Problem Solving

## EDUCATION

**University of Engineering and Technology Lahore** — Lahore, Pakistan

*Bachelor of Science in Mechatronics and Control Engineering* — Sep 2021 – June 2025

CGPA of 3.1/4.0

*Awards: - **MicroPython Trainer** (GC University Lahore); **Brain of Lahore** (UET)*

*Courses: - Control Systems, Modelling and Analysis of the Systems, Robotics, Embedded Systems, Industrial Automation, Design of Machine Elements (CAD), Intelligent Systems, Instrumentation and Measurements, Mobile Robotics*

## EXPERIENCE

**Punjab Information Technology Board (PITB)**

Model Town, Lahore, Pakistan

*Associate Embedded Systems Engineer* — Oct 2025 – Present

- Designed and implemented embedded systems solutions for government and private-sector projects, including Driver Console Units and **OBD-II analyzers** for **electric, SPEEDO, and METRO** bus diagnostics and monitoring.
- Collaborated with cross-functional software teams to integrate **C/C++ firmware with higher-level applications (Android/NDK)**, participated in code reviews, and produced detailed technical documentation

**NEWAY Automobile Industries** — Lahore, Pakistan

*Junior Automation Engineer (Part Time)* — May 2025 – Present

- Designed and implemented PLC-based automation systems for automotive manufacturing, including control panel design, field device integration, and programming/troubleshooting using Siemens (TIA Portal), Allen-Bradley (Studio 5000), and WECON PLCs.
- Conducted R&D on mobile robots with human–robot interaction and implemented IIoT solutions for machine data acquisition, server integration, operating-hour monitoring, and energy cost estimation.

**ALBA Engineering Company** — Daska, Pakistan

*Intern* — July 2024 – Aug 2024

- Developed and implemented deep learning–based parts inspection techniques to monitor component condition and predict remaining useful life, enhancing preventive maintenance and quality control.
- Proposed a robotic system for bulk part handling, performing **forward and inverse kinematics** calculations, and validating the design through MATLAB simulations to ensure accurate pick-and-place operations.

**Sports Industries and Development Center Sialkot (SIDC)** — Daska, Pakistan

*Intern* — June 2023- July 2023

- Designed and implemented a prototype for a fully automated **nozzle-hole making machine for footballs**, using a stepper motor and TI TIVA TM4C123GH6PM MCU in a closed-loop control system, enabling dual-hole production compared to the single-hole capability of the previous machine.

## PROJECTS

- **Design and Manufacture the Student Formula1 car:** Designed the Formula1 student race car from scratch, including chassis design and fabrication, Powertrain Design, Tractive Control System Design according to the FSAE Rules Germany.
- **Design and Fabrication of 5 Bar 2-DOF Robot:** Designed and fabricated a 5-Bar 2-DOF robot with stepper motor actuation, and a well programmed in ATMEL MCU, performing **forward and inverse kinematics** in MATLAB, and implementing the mechanical design in SolidWorks.
- **Modelling and Simulation of KUKA KR5 Robot:** Designed a robot in SolidWorks and imported it into MATLAB to implement the forward and inverse kinematics.
- **Modelling and Simulation of UR16e and ABB IRB 2600 Robots:** Modeled and simulated UR16e and ABB IRB 2600 robots in MATLAB, implementing end-effector motion along a square trajectory.